

In my role as a Data Management and Ethics Analyst in a Business Unit Data Office (BUDO), I am responsible for metadata management, strategic data quality, and data issue resolution. The Chief Data and Analytics Office defines the data strategy, while BUDOs ensure its implementation. I also support data product governance and the management of data risks and controls. My work involves collaboration with colleagues such as data issue managers, data governance and privacy specialists, as well as stakeholders across various business areas, including data and software engineers.

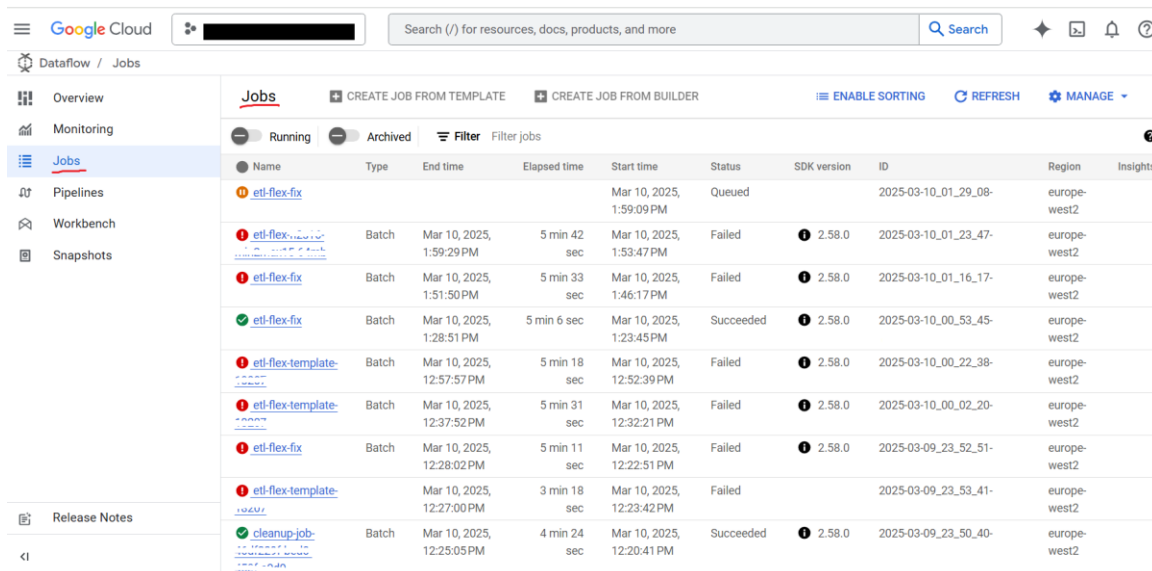
Monitoring Data Stores for Performance and Availability (K1, S7)

The company operates 16 distinct brands and provides various financial services – mortgages, credit cards, current accounts, pensions, investments, car financing, insurance, loans, and more – serving over 26 million customers. This generates and consumes vast amounts of data. Each product area manages applications that collect, store, and process data. For instance, Mortgage Sales Data (MSP) is a system of record for new mortgage applications. After approval, income and affordability data are archived, while personal customer data is transferred to a separate system for existing customers.

The main data stores include relational databases (Teradata, BigQuery, PostgreSQL) and NoSQL databases (MongoDB, Firestore, Apache Cassandra, HBase, Google Bigtable, Neo4J), along with private and public cloud platforms (GCP, AWS, Azure, Oracle). Google Cloud Platform is the strategic cloud provider, with ongoing migration from on-premises and legacy systems. Cloud Dataflow is a managed stream and batch processing service. It is a core component for building data pipelines that allows data transformation and processing. Monitoring of Dataflow jobs is possible in the Dataflow Jobs section in the GCP console. Here a data engineer can see a job ID, status (Succeeded/Failed), and execution time. As shown in Fig.1, there is a red warning sign next to failed jobs, making it easier to quickly identify them. An unusually high elapsed time can also indicate an issue, especially if a job has been running for a long time and has not finished yet. In this case, a data engineer can manually cancel it and explore the logs further. Clicking on the job name brings you to the execution graph, where you can investigate each step in detail (Fig. 2). As demonstrated in Fig. 3, error messages and system warnings are also available. For example, incompliance with the destination schema (wrong data type or a null value in a mandatory field) can result in job failure.

Migrating from a legacy data warehouse to BigQuery has already brought a number of benefits. Metadata capture improved data discoverability; an intuitive, easy-to-use

interface promotes fast learning; query speed has improved from a few minutes to a few milliseconds, especially when using clustering and partitioning. However, ignoring GCP best practices and query optimisation techniques, such as querying only required columns, query by partition date, using the cache and materialised views, optimising joins, may result in higher compute costs.



Name	Type	End time	Elapsed time	Start time	Status	SDK version	ID	Region	Insights
eti-flex-fix				Mar 10, 2025, 1:59:09 PM	Queued		2025-03-10_01_29_08-	europa-west2	
eti-flex-fix	Batch	Mar 10, 2025, 1:59:29 PM	5 min 42 sec	Mar 10, 2025, 1:53:47 PM	Failed	2.58.0	2025-03-10_01_23_47-	europa-west2	
eti-flex-fix	Batch	Mar 10, 2025, 1:51:50 PM	5 min 33 sec	Mar 10, 2025, 1:46:17 PM	Failed	2.58.0	2025-03-10_01_16_17-	europa-west2	
eti-flex-fix	Batch	Mar 10, 2025, 1:28:51 PM	5 min 6 sec	Mar 10, 2025, 1:23:45 PM	Succeeded	2.58.0	2025-03-10_00_53_45-	europa-west2	
eti-flex-template	Batch	Mar 10, 2025, 12:57:57 PM	5 min 18 sec	Mar 10, 2025, 12:52:39 PM	Failed	2.58.0	2025-03-10_00_22_38-	europa-west2	
eti-flex-template	Batch	Mar 10, 2025, 12:37:52 PM	5 min 31 sec	Mar 10, 2025, 12:32:21 PM	Failed	2.58.0	2025-03-10_00_02_20-	europa-west2	
eti-flex-fix	Batch	Mar 10, 2025, 12:28:02 PM	5 min 11 sec	Mar 10, 2025, 12:22:51 PM	Failed	2.58.0	2025-03-09_23_52_51-	europa-west2	
eti-flex-template		Mar 10, 2025, 12:27:00 PM	3 min 18 sec	Mar 10, 2025, 12:23:42 PM	Failed		2025-03-09_23_53_41-	europa-west2	
cleanup-job	Batch	Mar 10, 2025, 12:25:05 PM	4 min 24 sec	Mar 10, 2025, 12:20:41 PM	Succeeded	2.58.0	2025-03-09_23_50_40-	europa-west2	

Figure 1 – Dataflow jobs

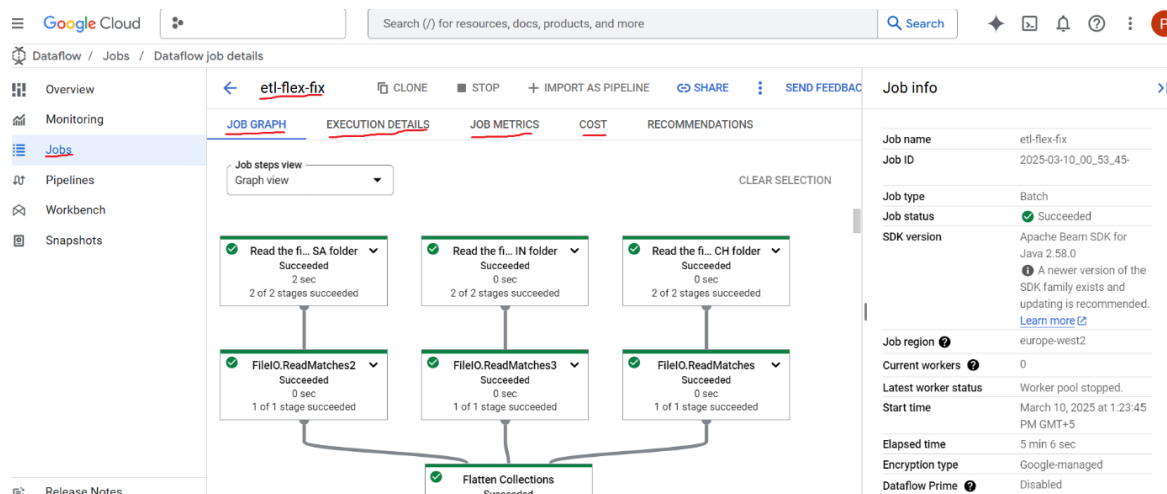


Figure 2 – Dataflow Execution Graph

Data Normalisation and Its Advantages (K3)

Data normalisation is a process of modelling and organising structured data into smaller tables to conform to normal forms. The data usually adhere to the first three normalisation principles, ensuring that each column contains only atomic values, eliminating redundant data, and removing transitive dependencies. A good example of this is reference data, i.e., the data that classifies and describes other data. It can be internally defined or

externally sourced and is typically slow-changing. For example, ISO country codes and corresponding countries and territories are standardised and used globally. By creating a reference ISO data product, the organisation ensures that the same data is used across all areas, it is always up to date and of appropriate quality. Transaction Classification System is an example of internally developed normalised reference data used to classify transactions in order to provide insights into the income and expenditure behaviour of customers as well as detect fraudulent transactions. As shown in Fig. 4, category_code is a unique identifier (PK), and the table contains a top-level category and two subset categories.

The screenshot shows the Google Cloud Dataflow console for a job named 'etl-flex-fix'. The job is in a 'Failed' state. The logs section shows several error messages from workers, indicating a 'Workflow failed' due to a 'Wait/Map/ParMultiDo(Anonymous)' operation. The job information panel on the right provides details about the job, including its name, ID, type, status, SDK version, region, and other configuration details.

SEVERITY	TIMESTAMP	SUMMARY
Info	2025-03-10 19:42:44.711 IST	No older entries found matching current filter.
Error	2025-03-10 19:42:45.367 IST	Error message from worker: org.apache.beam.sdk.util.UserCodeException: jav...
Error	2025-03-10 19:42:45.978 IST	Error message from worker: org.apache.beam.sdk.util.UserCodeException: jav...
Error	2025-03-10 19:42:46.558 IST	Error message from worker: org.apache.beam.sdk.util.UserCodeException: jav...
Error	2025-03-10 19:42:46.944 IST	Workflow failed. Causes: S71:Wait/Map/ParMultiDo(Anonymous)...
Info	2025-03-10 19:42:47.000 IST	No newer entries found matching current filter.

Job name	etl-flex-fix
Job ID	2025-03-10_07_08_39-
Job type	Batch
Job status	Failed
SDK version	Apache Beam SDK for Java 2.58.0 A newer version of the SDK family exists and updating is recommended. Learn more
Job region	eu-west-2
Current workers	0
Latest worker status	Worker pool stopped.
Start time	March 10, 2025 at 7:38:39 PM GMT+5
Elapsed time	5 min 27 sec
Encryption type	Google-managed
Dataflow Prime	Disabled
Runner v2	Enabled

Figure 3 – Dataflow Logs

CATGRY_CI	CATGRY_L1	CATGRY_L2	CATGRY_L3
0	Benefits	Testing Allowance	Other
1	Benefits	Attendance Allowance	Other
2	Benefits	Carers Allowance	Other
3	Benefits	Carers Support	Other
4	Benefits	Child Benefit	Other
5	Benefits	Disability Living Allowance	Other
6	Benefits	Education Maintenance Allowance	Other
7	Benefits	Housing Benefit	Other
8	Benefits	Incapacity Benefit	Other
9	Benefits	Income Support	Other
10	Benefits	Independent Living	Other
11	Benefits	Industrial Injuries Benefit	Other
12	Benefits	Job Seekers Allowance	Other
13	Benefits	Other	Other
14	Benefits	Social Fund	Other
15	Benefits	State Pension	Other
16	Benefits	Victim Support	Other
17	Benefits	Winter Fuel Payment	Other
18	Benefits	Working Families Tax Credit	Other
393	Benefits	Universal Credit	Other
395	Benefits	Employment and Support Allowance	Other
401	Business To Business	Accounts Paid	Other
19	Communication	Internet	Other
20	Communication	Mobile Phone	Other
21	Communication	Multi Media Service	Other
22	Communication	Other	Other
23	Financial Services	Assurance	Other
24	Financial Services	Cards	Balance Transfer
25	Financial Services	Cards	Clubcard
26	Financial Services	Cards	Other
27	Financial Services	Cards	Storecard

Figure 4 – A snapshot of Transaction Classification System

Data Quality and Risk Management (K5, K10)

The inherent risks associated with working with data include ineffective processing of core data; ineffective data privacy and ethics process execution; ineffective data accountability, decision making and documentation; failure to effectively govern, manage, and control core data processed by third-party suppliers and their sub-contractors. It can affect quality, reliability, retention and privacy of data and records, resulting in poor customer and colleague outcomes, increased operational costs, poor decision making, reputational damage, customer and colleague detriment, and regulatory sanctions. Data risks and controls are recorded on RCSA, which is a Risk and Control Self-Assessment system. The tool also includes information about risk owners, inherent and residual risk assessment details, control effectiveness, and any associated assessment plans (see Fig. 5).

GDPR (General Data Protection Regulation) is a European Union regulation on data protection and privacy for all individuals within the EU, EEA, and UK. Its key aspects are data protection, rights of data subjects (DSAR), data controllers and processors, and compliance and enforcement. Data privacy teams ensure that only authorised users have access to confidential data, and any breaches are recorded and reported to regulatory authorities. Data architects ensure that data products are designed and built to facilitate data

subject access requests (DSAR). Customers should be informed how their personal data is used, including the use of AI. Personal data should be used only for the intended use case specified in Privacy & Records Impact Assessment (PRIA).

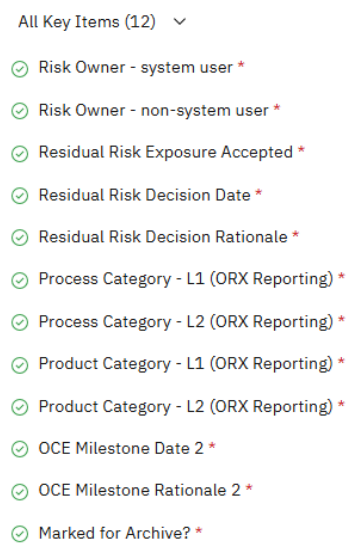
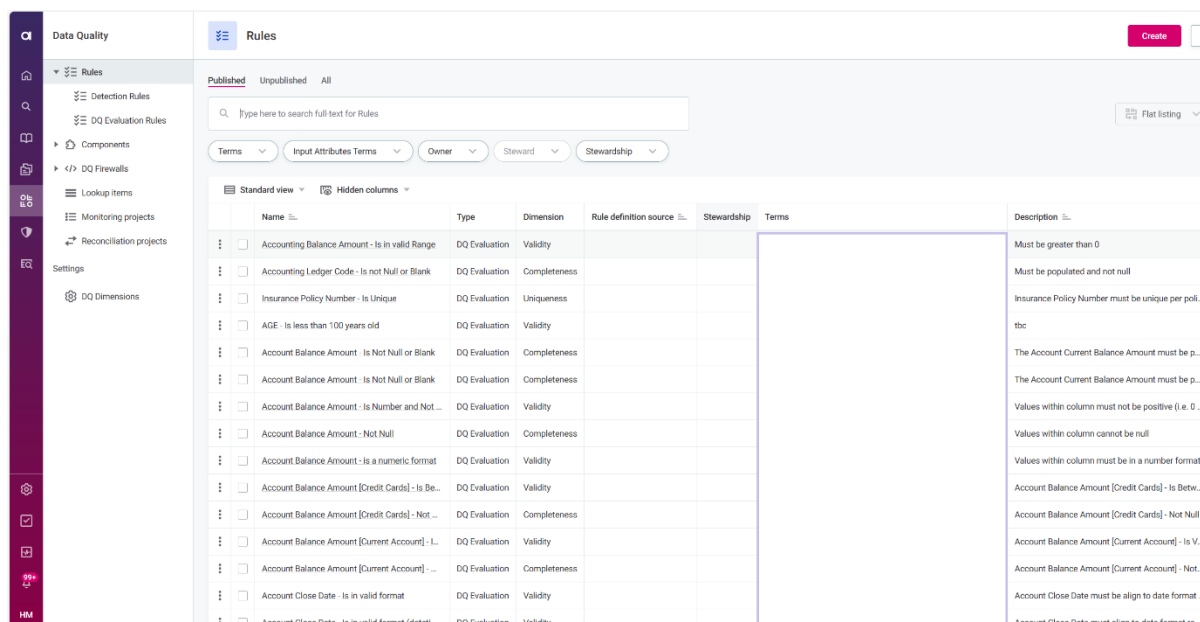


Figure 5 – RCSA Sections

Ataccama, shown in Fig. 6, is a strategic data quality software in the organisation. Its main capabilities are data profiling, rules creation, and dashboards creation. The drawbacks include licence costs, vendor lock-in, bugs, and frequent upgrades that include radical web interface and functionality changes. The rules are monitored and run regularly within agreed time periods, e.g. monthly. Monitoring results are displayed on the dashboard that includes the number of passed and failed fields in the latest run as well as data quality over time. A call with Data Owner is scheduled monthly to present the dashboard and log any data issues on Collibra for future remedy. Data profiling is performed for any new data or when a significant change in data took place. It is helpful for understanding the data and identifying what data quality rules are needed.

Any employee can raise a data issue on Collibra, a data management and quality repository. Based on the application where the issue is identified, it is passported to the relevant BUDO. For example, I did initial investigation and triage for an issue related to missing middle names on new mortgage applications. After talking to application owners and data engineers, I found out that when existing customers submit a new mortgage application online, they sometimes do not mention their middle name as it is an optional field. The data from the mortgage system flows into our customer data system, and updates the records. If we already had customer's middle name but the customer did not mention it in a new mortgage application, the record is overwritten in the customer data system and the middle

name is dropped. Since the data is used for all other products, as well as by the financial crime and sanctions team, this behaviour poses a risk of degrading the quality of customer personal data, resulting in longer wait periods for the mortgage approval, sanctions and regulatory costs, inaccurate credit score and affordability checks, etc.



The screenshot shows the Ataccama Rules interface. On the left is a sidebar with navigation options: Data Quality, Rules, Detection Rules, DQ Evaluation Rules, Components, DQ Firewalls, Lookup Items, Monitoring projects, Reconciliation projects, Settings, and DQ Dimensions. The main area is titled 'Rules' and has tabs for 'Published' and 'Unpublished'. Below the tabs is a search bar and several filter dropdowns: Terms, Input Attributes Terms, Owner, Steward, and Stewardship. A 'Flat listing' button is on the right. The main table displays a list of rules with columns: Name, Type, Dimension, Rule definition source, Stewardship, Terms, and Description. The table is currently in 'Standard view' and shows 15 rules.

Name	Type	Dimension	Rule definition source	Stewardship	Terms	Description
Accounting Balance Amount - Is in valid Range	DQ Evaluation	Validity				Must be greater than 0
Accounting Ledger Code - Is not Null or Blank	DQ Evaluation	Completeness				Must be populated and not null
Insurance Policy Number - Is Unique	DQ Evaluation	Uniqueness				Insurance Policy Number must be unique per poli...
AGE - Is less than 100 years old	DQ Evaluation	Validity				tbc
Account Balance Amount - Is Not Null or Blank	DQ Evaluation	Completeness				The Account Current Balance Amount must be p...
Account Balance Amount - Is Not Null or Blank	DQ Evaluation	Completeness				The Account Current Balance Amount must be p...
Account Balance Amount - Is Number and Not ...	DQ Evaluation	Validity				Values within column must not be positive (i.e. 0 ...
Account Balance Amount - Not Null	DQ Evaluation	Completeness				Values within column cannot be null
Account Balance Amount - Is a numeric format	DQ Evaluation	Validity				Values within column must be in a number format
Account Balance Amount [Credit Cards] - Is Be...	DQ Evaluation	Validity				Account Balance Amount [Credit Cards] - Is Betw...
Account Balance Amount [Credit Cards] - Not ...	DQ Evaluation	Completeness				Account Balance Amount [Credit Cards] - Not Null
Account Balance Amount [Current Account] - L...	DQ Evaluation	Validity				Account Balance Amount [Current Account] - Is V...
Account Balance Amount [Current Account] - ...	DQ Evaluation	Completeness				Account Balance Amount [Current Account] - Not...
Account Close Date - Is in valid format	DQ Evaluation	Validity				Account Close Date must be align to date format ...
Account Close Date - Is in valid format (dateti...	DQ Evaluation	Validity				Account Close Date must align to date format re...

Figure 6 – Ataccama rules

Ingestion Frameworks and Optimisation (K18, S15)

As the company migrates its data to the cloud, creating data products on GCP is currently the main focus for data engineers, architects, and governance specialists. The proposed typology includes three kinds of data products: origin, foundation, and consumption. Origin data products are complete replicas of the data and processes that previously existed on on-premises systems. Streaming data pipelines are used to receive data in real or near-real time. These pipelines require low latency and often involve high volumes of data transmitted simultaneously. For example, the Transport team receives near-real-time data from third parties using Kafka. This data includes CAP codes, which are unique identifiers that detail the genetic makeup of a vehicle, including its make, model, engine, trim, and other features. The bank needs this data for real-time vehicle valuations and risk monitoring when financing a particular vehicle.

A use case for on-demand services is calculating a customer's affordability for a specific product. For instance, when a customer enters their details on the website to see how much they may be able to borrow when applying for a new credit card, the decision is based on their credit score, other products they hold with the bank, income, and spending patterns.

Interacting with the website and completing the application form triggers relevant data sources in real time, so the calculations are generated on demand.

Lastly, an example of batch processing is the migration of historical data from legacy systems to Google Cloud. Files in CSV format land in a Cloud Storage bucket. The arrival of these files triggers Cloud Functions, which perform a file validation operation. Dataflow is used for schema validation, de-duplication, and the removal of any bad data, while Cloud Composer orchestrates the pipelines. Valid data is then loaded into BigQuery, while invalid data is written to a separate BigQuery dataset for further analysis (see Fig. 7). Metadata and lineage are also collected via Dataplex, while Log Analytics and Audit Control ensure that all activities are captured and stored for debugging, monitoring, and analysis.

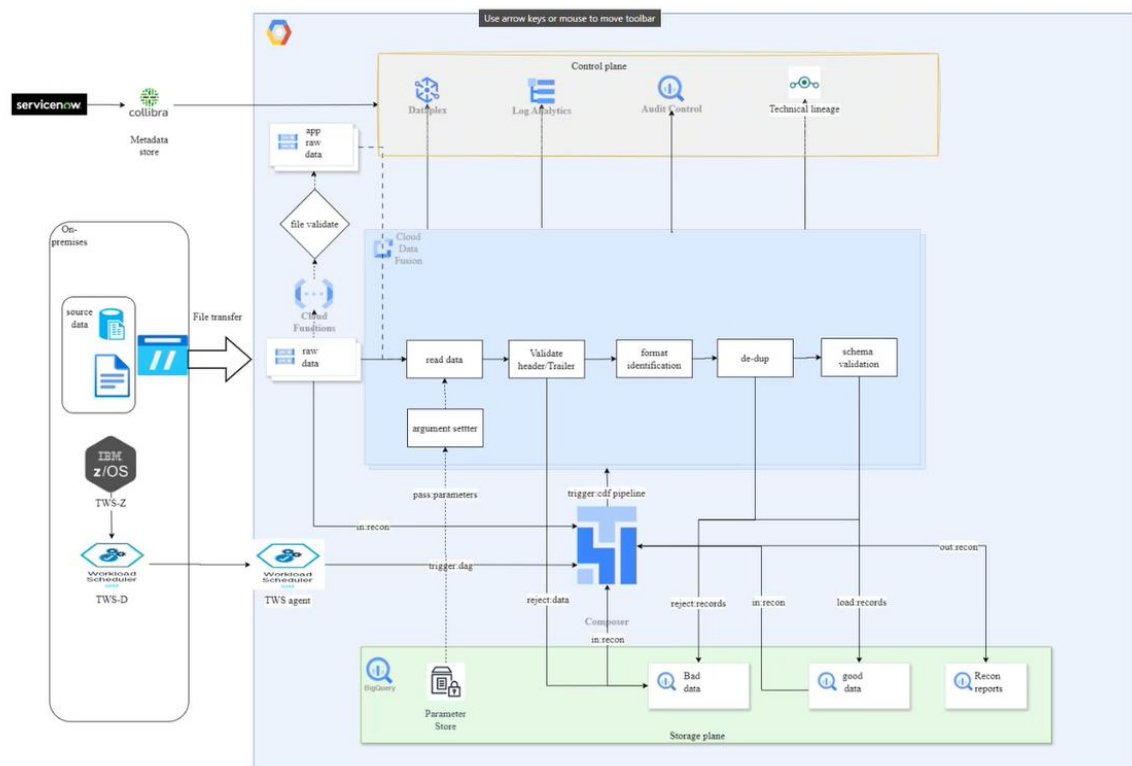


Figure 7 – Batch data processing on Google Cloud